

Should E-Cigarette Regulation Be Based on Randomized Trials or Observational Studies?

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8 See also Fairchild, p. 221, Gee et al., p. 224, Grant, p. 227, and Wang et al., p. 230.

In a systematic review, Wang et al. (p. 230) considered several types of evidence about the effects of electronic cigarettes (e-cigarettes) on smoking cessation.

RANDOMIZED CLINICAL TRIAL EVIDENCE

Synthesizing the results of randomized clinical trials (RCTs) using meta-analysis, Wang et al. found that providing free e-cigarettes probably helps some people quit smoking. This is consistent with a recent Cochrane review that found that free e-cigarettes provided in trials were more effective than nicotine replacement for smoking cessation and possibly more effective than behavioral therapy.¹ Both reviews highlight important clinical differences across individual trials, and both reviews show that e-cigarettes have small absolute benefits because most people are unsuccessful in their attempts to quit smoking to tobacco. Moreover, both reviews point to concerning evidence that people use e-cigarettes longer than other smoking

cessation aids. In the long-term, switching to e-cigarettes might be better than continuing to smoke tobacco; however, these trials do not show whether the short-term benefits of e-cigarettes offset their long-term harms when compared with nicotine replacement, such as gums or lozenges, which are effective for smoking cessation and used for a shorter time.²

Wang et al. argue that if regulators such as the US Food and Drug Administration had approached e-cigarettes as they approach drugs and medical devices, RCTs restricted to people who are motivated to quit smoking would have been appropriate for assessing their potential benefits. Although their effects on smoking cessation are small, e-cigarettes might even compare favorably with other smoking cessation therapies such as bupropion, varenicline, and nicotine inhalers.

OBSERVATIONAL STUDIES

Because e-cigarettes have been regulated and sold as consumer products,

Wang et al. argue, observational studies are important to assess their effectiveness in the general population, which includes nonsmokers and people who are not motivated to quit smoking. As consumer products, e-cigarettes might do more harm than good if they increase nicotine use among people who smoke and are not motivated to quit, or they might do harm if they attract new nicotine users. E-cigarettes could theoretically increase tobacco use by increasing nicotine initiation among young people who later add tobacco cigarettes or switch from e-cigarettes to tobacco cigarettes. In an accompanying editorial, Gee et al. (p. 224) highlight why concerns about young people are central to the debate about the regulation and distribution of e-cigarettes.

To help understand the effects of e-cigarettes as consumer products, Wang et al. analyzed observational studies of the association between e-cigarettes and smoking cessation using two approaches. They conducted one analysis restricted to people who smoke and express motivation to quit smoking. In another analysis they included all people who smoke, regardless of motivation to quit smoking tobacco. In an accompanying editorial, Fairchild (p. 221) explains that guideline development groups have similarly grappled with the difficulty of answering closely related research questions for which different types of evidence about e-cigarettes are relevant.

CONFLICTING RESULTS?

At first glance, results from observational studies appear to differ from the results of RCTs. As Grant (p. 227) notes, we tend to have less certainty in the observational evidence compared with evidence from RCTs. Moreover, these analyses include multiple effect

estimates from some observational studies. Wang et al. used novel methods to account for the correlation between multiple estimates from the same study, and readers should interpret the consistency of these effects in light of the relationships among them.

RCTs suggest that e-cigarettes offer modest benefits, yet observational studies suggest that e-cigarettes are associated with less smoking cessation among people who express motivation to quit smoking tobacco. Considering the broader population of all people who smoke, regardless of motivation to quit, Wang et al. found that longitudinal studies indicate that e-cigarettes are, at best, modestly associated with smoking cessation. By contrast, summary effects from cross-sectional studies would be consistent with no important difference or a small decrease in smoking cessation among all people who smoke. However, results of these observational studies are so heterogeneous that the summary effects should be interpreted cautiously.

Looking deeper into their results, Wang et al. found qualitatively different associations for intense users and nonintense users of e-cigarettes. That is, intense e-cigarette use was associated with greater likelihood of smoking cessation, whereas less intense e-cigarette use was associated with lower likelihood of smoking cessation. These findings could highlight an important source of clinical heterogeneity—some people use e-cigarettes to quit smoking tobacco whereas other people use e-cigarettes alongside tobacco cigarettes—yet heterogeneity remained even within groups stratified by intensity.

Wang et al. argue that, as consumer products in the United States, e-cigarettes do not meet the standard for the “protection of public health” because it is unclear whether they increase

smoking cessation across the population. Gee et al. concur, whereas Fairchild (p. 221) and others interpret the evidence differently, concluding that the profound and well-known harms of smoking tobacco justify widespread distribution of e-cigarettes. Grant suggests that closer inspection of the observational evidence might have allowed Wang et al. to say more about certainty in this body of evidence and that greater transparency would have strengthened the review.

POLICY IMPLICATIONS

The review by Wang et al. focuses on smoking cessation and does evaluate whether “dual use” in the general population might decrease harm by reducing overall tobacco use. As Fairchild notes, observational studies of current smokers also omit people who used e-cigarettes successfully to quit smoking tobacco. Moreover, Wang et al. do not address ongoing debates about nicotine initiation, particularly in young people. Nonetheless, Wang et al. make a case for rethinking the regulatory approach to e-cigarettes.

Even if e-cigarettes do reduce tobacco smoking in the general population, the public health benefits of e-cigarettes might be separable from their availability as consumer products. Perhaps e-cigarettes must be sold in places such as supermarkets and gas stations to reach all people who wish to try them; however, other methods of distribution might reach the target group of adults who are motivated to stop or reduce smoking tobacco. Wang et al. argue that e-cigarettes could be made available by prescription, like nicotine inhalers. They explain that most e-cigarette users in the general population are not daily e-cigarette users, and the latter group accounts for modest increases in smoking

cessation in both RCTs and observational studies. Greater efforts to limit e-cigarette availability to adults who are ready to replace tobacco with e-cigarettes, the target population for whom there is rigorous evidence that their benefits likely outweigh their harms, could realize the benefits for this target group while minimizing harm to the rest of the population. **AJPH**

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CONFLICTS OF INTEREST

I have no conflicts of interest to declare.

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